Causal Evaluation Hierarchy

Strict dependency: success at Level N requires Level $N-1$

Increasing Causal Depth

L1 State Identification

Interpreting current machine states and detecting anomalies from sensor data.

L2 Intervention Reasoning

Predicting the immediate impact of control changes or faults at the present timestep.

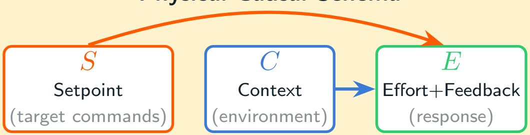
L3 Counterfactual Analysis

Simulating alternative historical timelines to assess hypothetical past conditions.

L4 Decision Support

Synthesizing multi-step recovery procedures using sensor fusion and technical manuals.

Physical Causal Schema



Healthy: $E = f(S, C)$ | Fault: $E \neq f(S, C)$

Fault ⇔ empirical deviation in *Effort* given *Setpoint* and *Context*

Scalable Q&A Generation Engine

+80 Expert Templates

Expert-authored across all 4 levels

Industrial Datasets

FactoryWave · AURSAD · Simulated Robotic

Real + simulated sensor streams

Dynamic Variable Sampling

Deterministic ground truth · Vendi Score validates diversity

Multi-Select | **Scalar** | **Tensor** | **Ranking** | **Free form**

Comparison with Existing Benchmarks

Benchmark	Robot	Multi. TS	Size	Counter.	Rank	Num.	Free	Novel TS
TimeSeriesExam	✗	✗	$\sim$700	✗	✓	✗	✗	✗
ChatTS	✗	✓	$\sim$134k	✗	✓	✓	✗	✗
EngineMT-QA	✓	✓	$\sim$110k	✗	✓	✓	✓	✗
TSAQA	Part.	✗	$\sim$210k	✗	✓	✗	✗	✗
Time-MQA	✗	✓	$\sim$200k	✗	✓	✓	✓	✗
MTBench	✗	✓	?	✗	✓	✓	✓	✗
QuAnTS	✗	✓	$\sim$150k	✗	✓	✓	✓	✗
CLEVR	✗	✗	$\sim$1M	✗	✗	✓	✗	✗
CLEVRER	✗	✗	$\sim$305k	✓	✗	✓	✗	✗
FactoryBench	✓	✓	**TBD**	✓	✓	✓	✓	✓

The only benchmark guaranteeing verifiable truth across deep causal temporal reasoning on machine data.